

## Training Configuration and Summary:

Dataset used: rssi\_odom\_dataset\_12apr25\_ajust.parquet

model\_type: LSTM  
group\_sec: 10  
timesteps\_orig: 60  
timesteps: 6  
batch\_size: 256  
min\_units: 32  
max\_units: 256  
units\_step: 32  
dropout\_min: 0.1  
dropout\_max: 0.3  
dropout\_step: 0.1  
activation\_functions: ['tanh', 'relu', 'elu']  
optimizers: ['adam', 'rmsprop', 'Nadam']  
min\_layers: 1  
max\_layers: 2  
max\_epochs: 150  
hyperband\_iterations: 3  
forecast\_steps: 20

### LSTM Multi-Step Training Report

Report Generated: 2025-05-10 13:58:49

Training Start: 2025-05-10\_13-05-53  
Training End: 2025-05-10\_13-12-21  
Training Duration: 6.46 minutes

Best Hyperparameters:  
Number of Layers: 1  
Activation: tanh  
Optimizer: rmsprop  
Dropout Rate: 0.20

MAE: 107757.3925  
MSE: 18915135915.9525  
RMSE: 137532.3086  
R2 Score: 0.8986

Multi-Step Forecast:  
Forecast Steps: 20  
MAE: 0.0094  
MSE: 0.0001  
RMSE: 0.0121  
R2 Score: 0.9636

Dataset size before down-sample: 85,269 rows  
Dataset size after down-sample: 85,007 rows

First 5 timestamps BEFORE:

1744397085760  
1744397085770  
1744397085789  
1744397085801  
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366  
1744464971377  
1744464971390  
1744464971402  
1744464971414

First 5 timestamps AFTER:

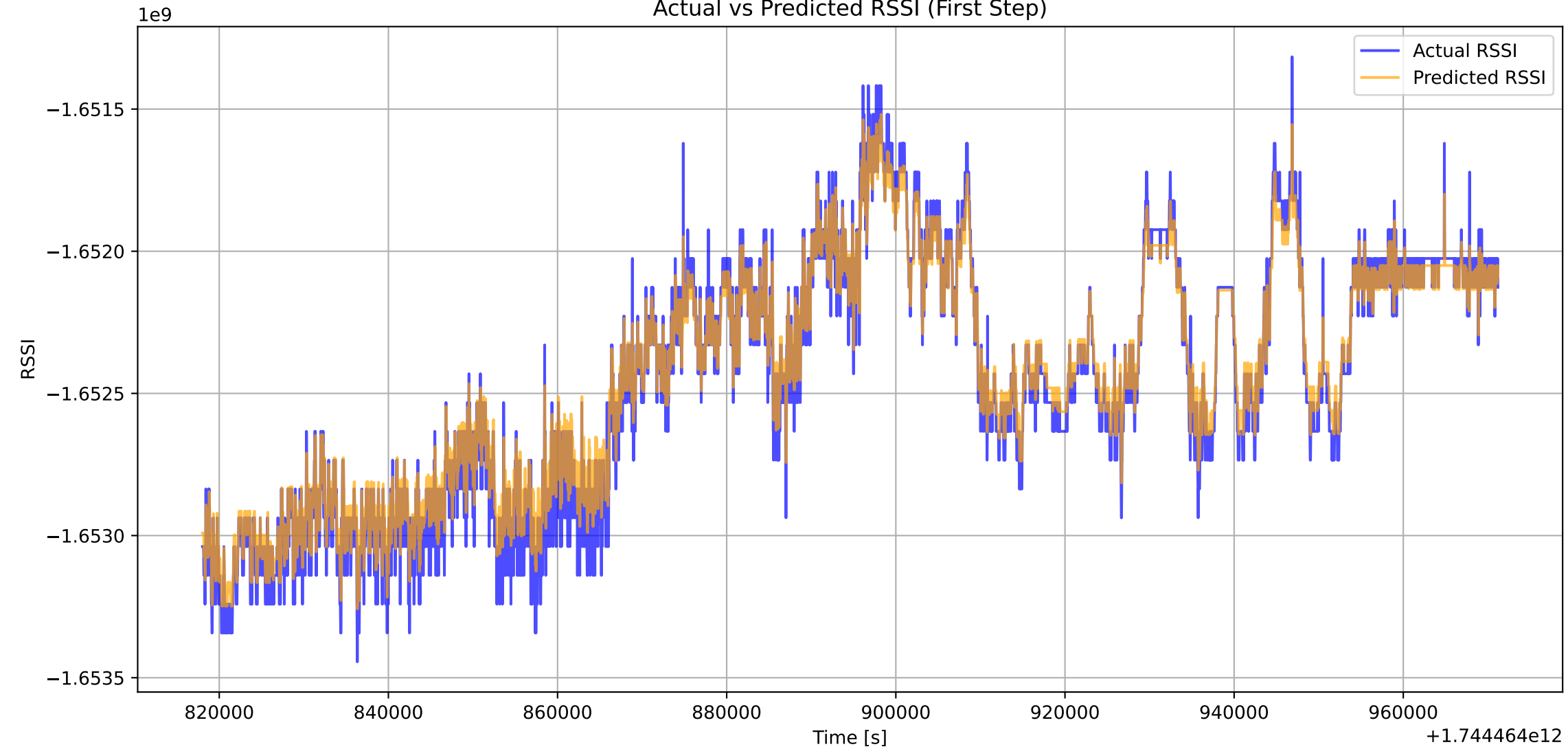
1744397085760  
1744397085770  
1744397085789  
1744397085801  
1744397085817

...

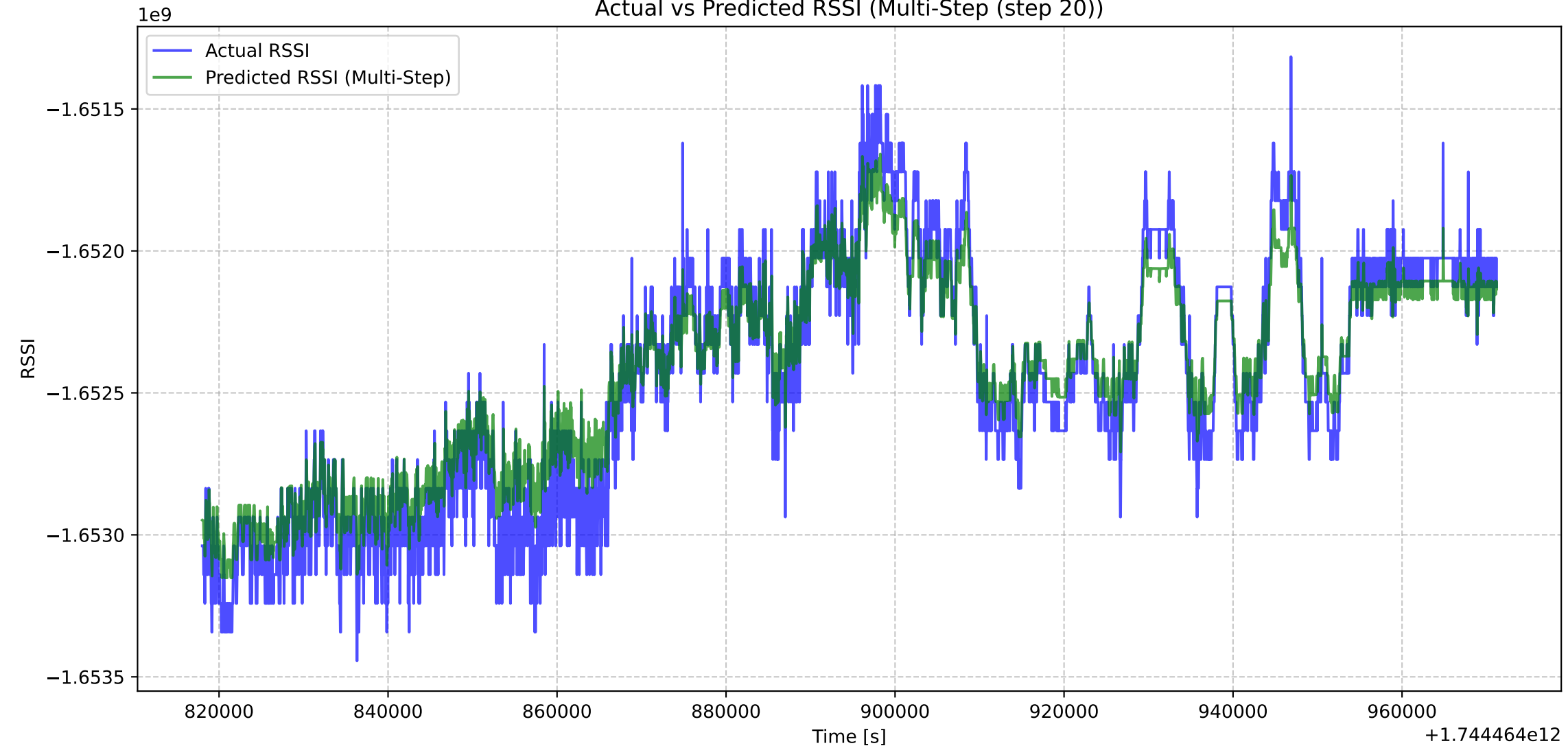
Last 5 timestamps AFTER:

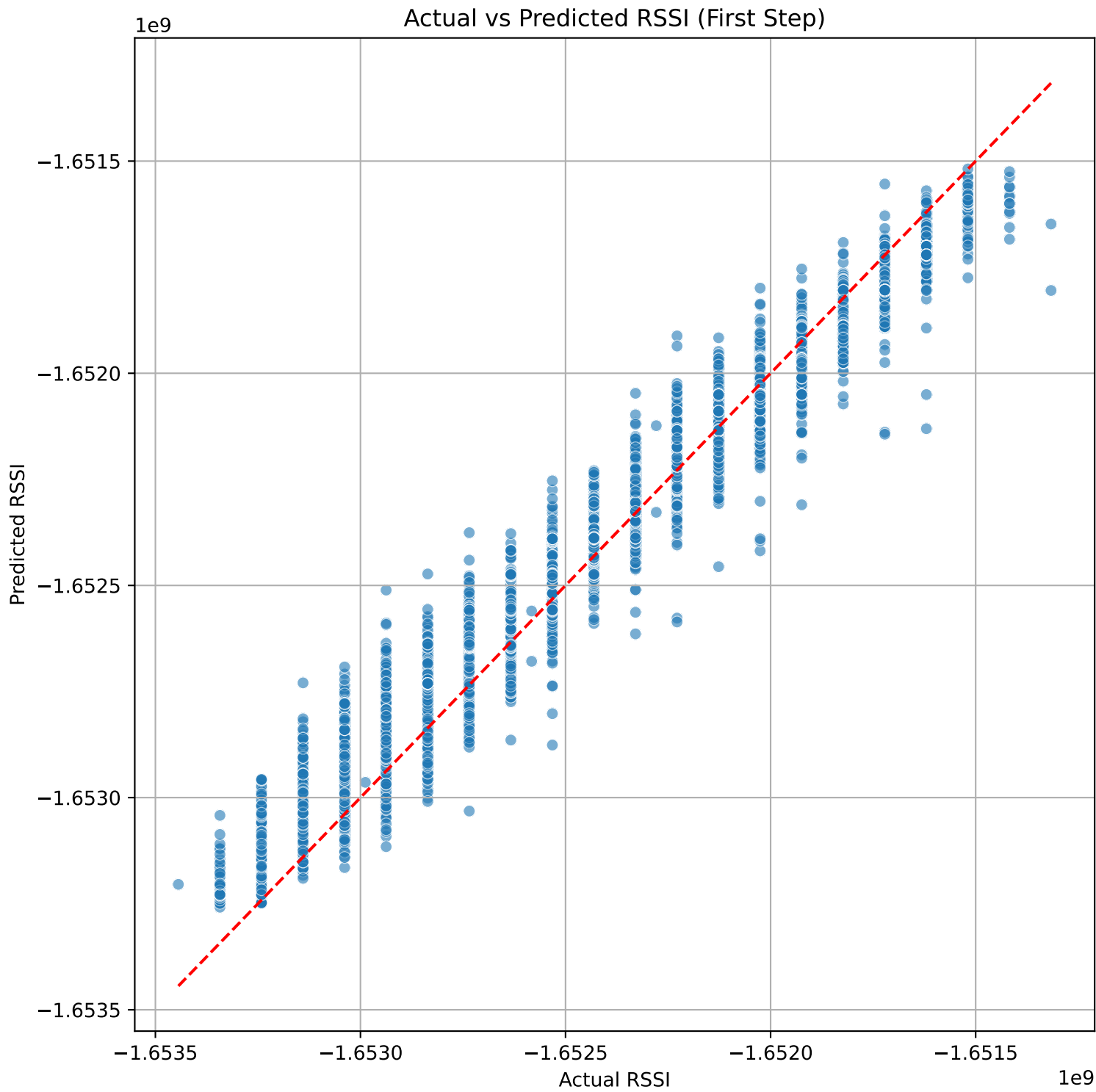
1744464971366  
1744464971377  
1744464971390  
1744464971402  
1744464971414

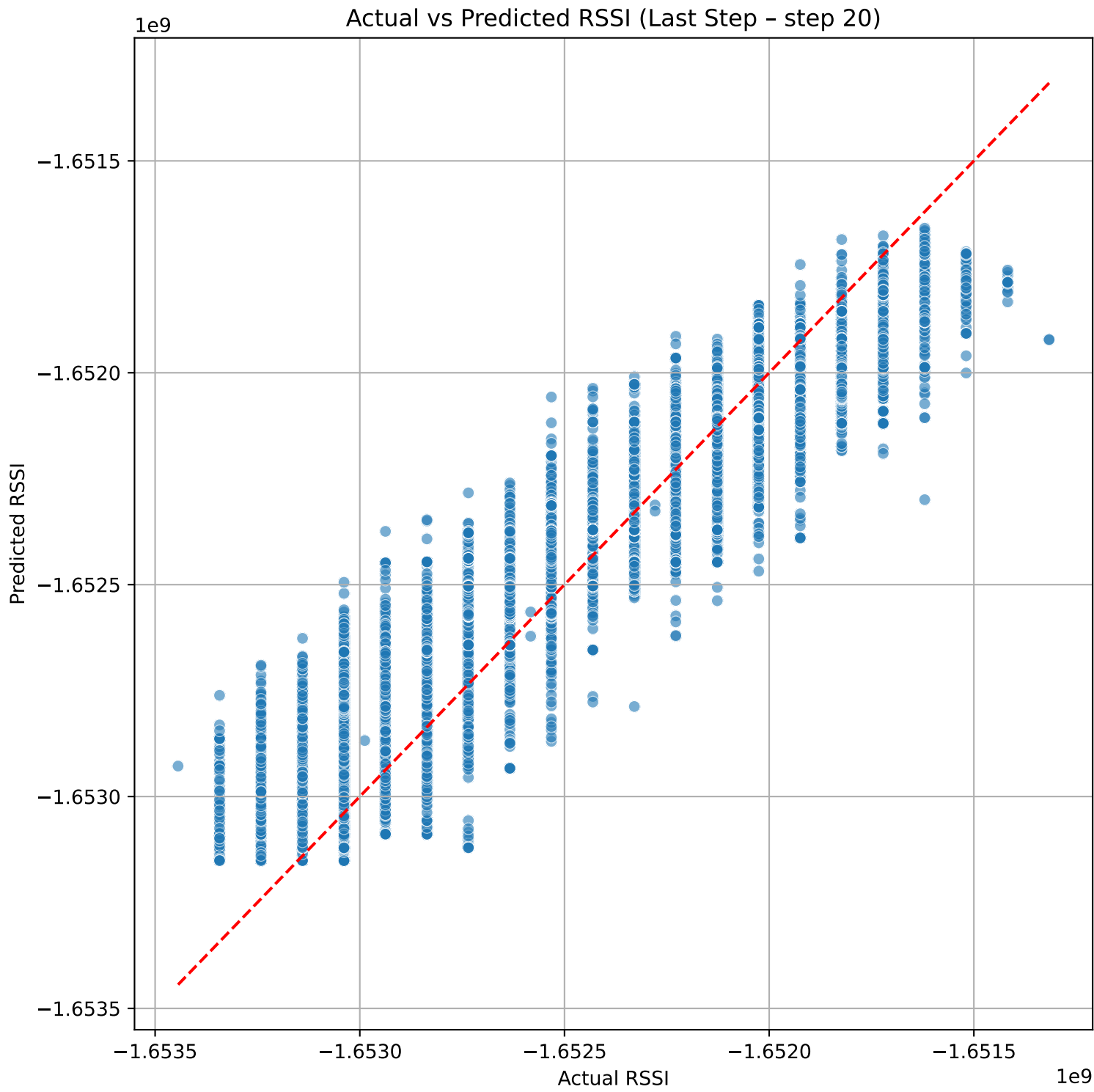
Actual vs Predicted RSSI (First Step)



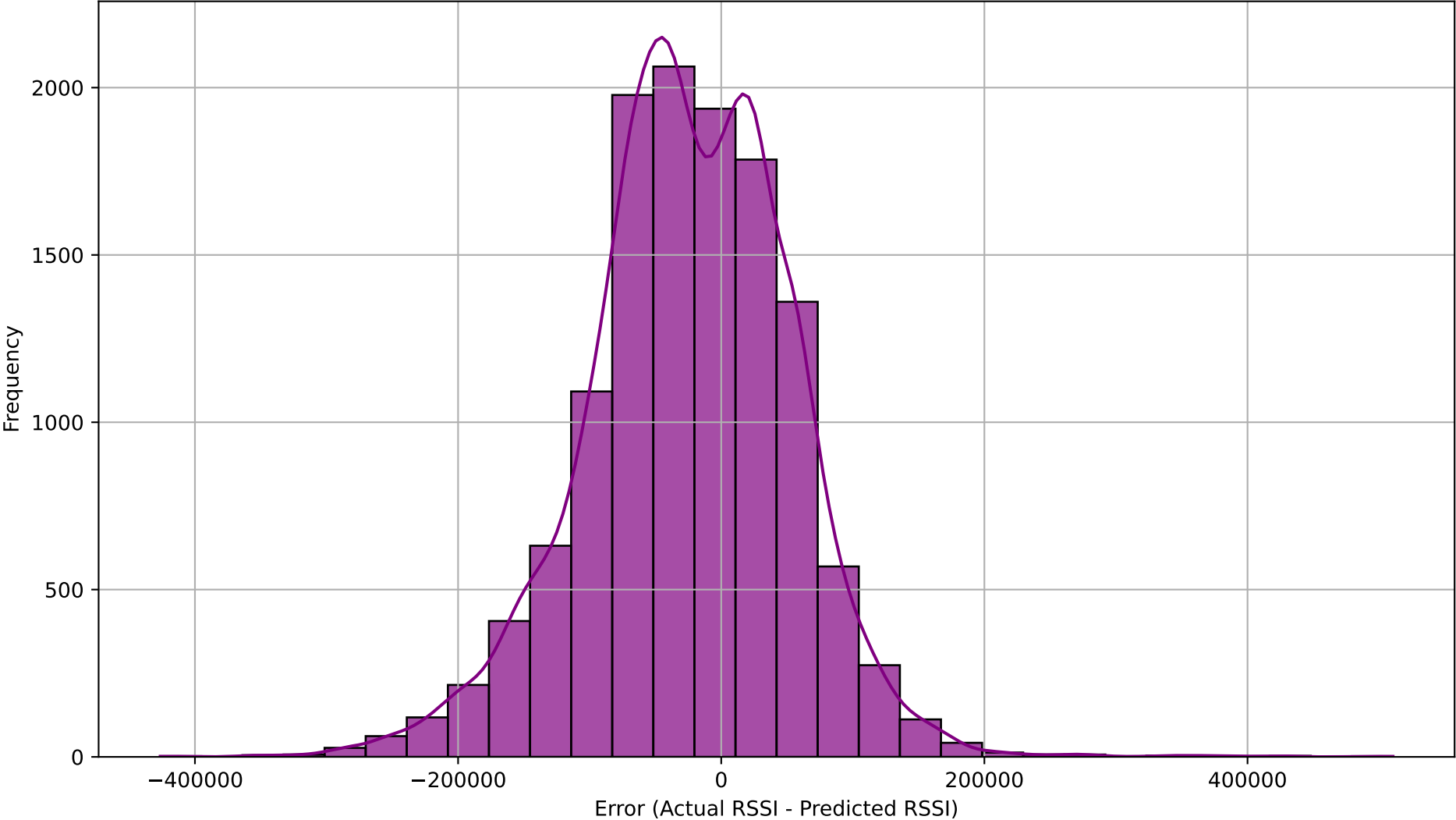
Actual vs Predicted RSSI (Multi-Step (step 20))



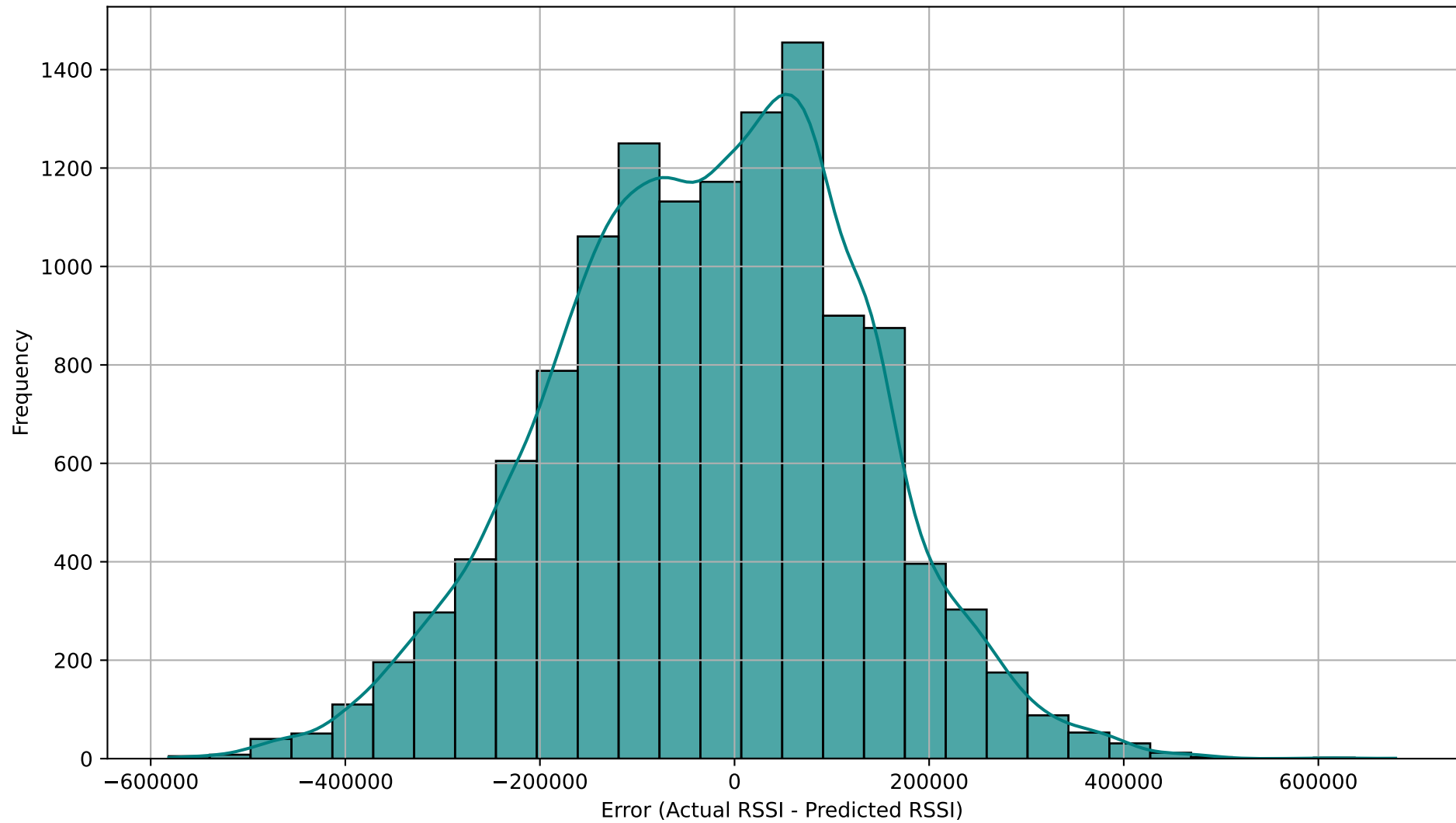




Distribution of Prediction Errors (First Step)

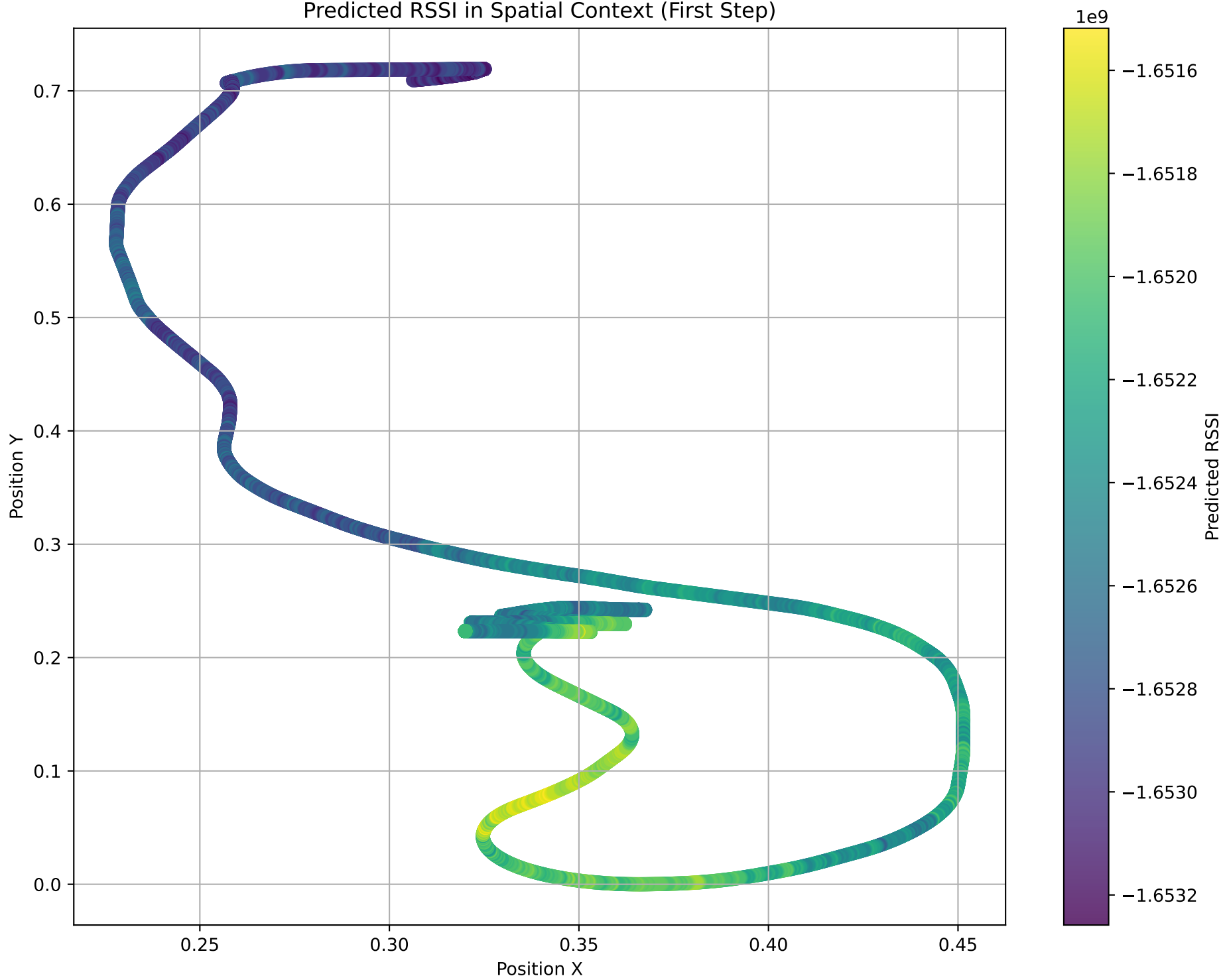


Distribution of Prediction Errors (Last Step - step 20)

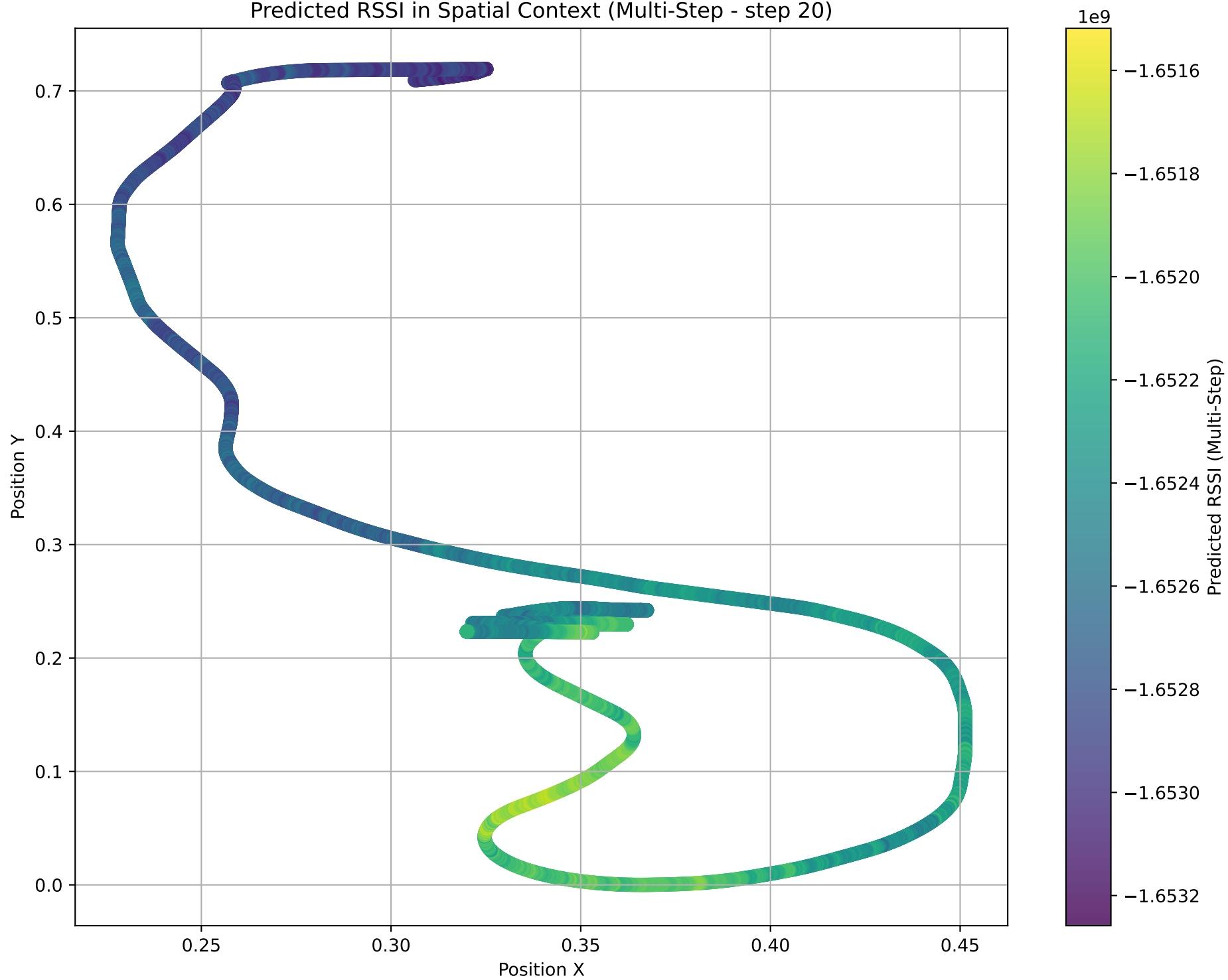




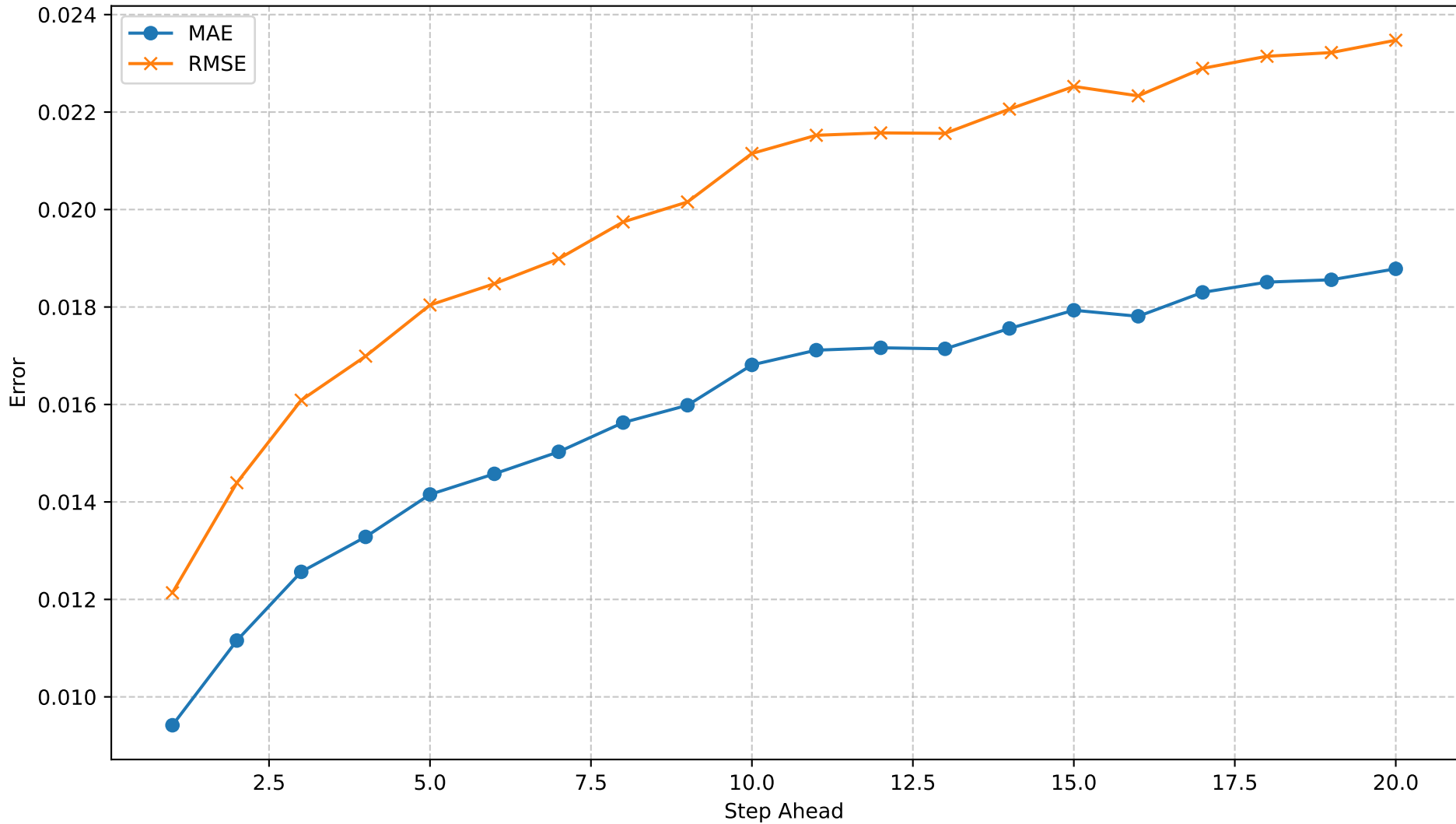
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



### Resumo Automático

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Dataset: rssi\_odom\_dataset\_12apr25\_ajust.parquet  
Modelo: LSTM  
Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.01 / 0.01
- Mediana : 0.02 / 0.02
- 20º passo : 0.02 / 0.02

Crescimento do erro: 99.4% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 20 (0.02), 19 (0.02), 18 (0.02)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.